Type MS · Factorising and solving a cubic

Answer 1 of 22 · Question 3 part b · 4 marks

2020 JAN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

3 (a)	$x = 4$ $64p - 496 + 100 + 12 = 0$ $64p = 384$ $p = 6$ *	M1 A1 (2)
(b)	$(x-4)(6x^2-7x-3)$ $(x-4)(2x-3)(3x+1)$ $x = 4, \frac{3}{2}, -\frac{1}{3}$	M1 A1 M1 A1 (4)
		[6]

Part	Mark	Additional Guidance
(a)	M1	For substituting $x = 4$ into the **given** expression, equating the expression = 0 **and** attempting to solve for p
	A1	For $p = 6$ * This is a show question so every step must be seen
(b)	M1	For attempting to divide $6x^3 - 31x^2 + 25x + 12$ by $(x-4)$ $\Rightarrow x-4\overline{)6x^3-31x^2+25x+12}$ $6x^2-7x+k$ (k is an integer)
	A1	For finding the correct 3TQ $6x^2 - 7x - 3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2-7x-3=(2x-3)(3x+1)$ Condone $\left(x-\frac{3}{2}\right)\left(x+\frac{1}{3}\right)$
	A1	For the correct solution seen: $x = 4.\ \frac{3}{2},\ -\frac{1}{2}$
	ALT – equates coefficients	
	M1	For stating $6x^3-31x^2+25x+12=(x-4)(Ax^2+Bx+C)\Rightarrow$ $6x^3-31x^2+25x+12=Ax^3+x^2(B-4A)+x(C-4B)-4C$ Minimum required is $A=6,\ B=-7$ and $C=k$
	A1	For $A=6,\ B=-7$ and $C=-3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2-7x-3=(2x-3)(3x+1)$
	A1	For the correct solution seen: $x = 4.\ \frac{3}{2},\ -\frac{1}{2}$
	ALT – by inspection	
	M1	For finding the **quadratic** factor minimum required is $[(x-4)](6x^2-7x+k)$
	A1	For finding the correct 3TQ $6x^2 - 7x - 3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2-7x-3=(2x-3)(3x+1)$
	A1	For the correct solution seen: $x = 4.\ \frac{3}{2},\ -\frac{1}{2}$
	Evidence of the 3TQ seen is required in part (b) $(x-4)(2x-3)(3x+1)=0\Rightarrow x=4,\ \frac{3}{2},\ -\frac{1}{3}$ **is M0**	

Type MS · Factorising and solving a cubic

Answer 1 of 22 · Question 3 part b · 4 marks

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Question number	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 1 of 22 · Question 3 part b · 4 marks

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Question number	Scheme	Marks
3 (a)	$x = 4$ $64p - 496 + 100 + 12 = 0$ $64p = 384$ $p = 6$ *	 M1 A1 (2)
(b)	$(x-4)(6x^2 - 7x - 3)$ $(x-4)(2x-3)(3x+1)$ $x = 4, \frac{3}{2}, -\frac{1}{3}$	M1 A1 M1 A1 (4)
		[6]

Part	Mark	Additional Guidance
(a)	M1	For substituting $x = 4$ into the **given** expression, equating the expression = 0 **and** attempting to solve for p
	A1	For $p = 6$ * This is a show question so every step must be seen
(b)	M1	For attempting to divide $6x^3 - 31x^2 + 25x + 12$ by $(x-4)$ $\Rightarrow x-4\overline{)6x^3 - 31x^2 + 25x + 12}$ $6x^2 - 7x + k$ (k is an integer)
	A1	For finding the correct 3TQ $6x^2 - 7x - 3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2 - 7x - 3 = (2x-3)(3x+1)$ | Condone $\left(x - \frac{3}{2}\right)\left(x + \frac{1}{3}\right)$
	A1	For the correct solution seen: $x = 4, \frac{3}{2}, -\frac{1}{2}$
	ALT – equates coefficients	
	M1	For stating $6x^3 - 31x^2 + 25x + 12 = (x-4)(Ax^2 + Bx + C) \Rightarrow$ $6x^3 - 31x^2 + 25x + 12 = Ax^3 + x^2(B - 4A) + x(C - 4B) - 4C$ Minimum required is $A = 6$, $B = -7$ and $C = k$
	A1	For $A = 6$, $B = -7$ and $C = -3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2 - 7x - 3 = (2x-3)(3x+1)$
	A1	For the correct solution seen: $x = 4, \frac{3}{2}, -\frac{1}{2}$
	ALT – by inspection	
	M1	For finding the **quadratic** factor minimum required is $[(x-4)](6x^2 - 7x + k)$
	A1	For finding the correct 3TQ $6x^2 - 7x - 3$
	dM1	For an attempt to factorise their 3TQ to give $6x^2 - 7x - 3 = (2x-3)(3x+1)$
	A1	For the correct solution seen: $x = 4, \frac{3}{2}, -\frac{1}{2}$
	Evidence of the 3TQ seen is required in part (b)	
	$(x-4)(2x-3)(3x+1) = 0 \Rightarrow x = 4, \frac{3}{2}, -\frac{1}{3}$ **is M0**	

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Answer 1 of 22 · Question 3 part b · 4 marks

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Question number	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 2 of 22 · Question 9 part b.ii · 3 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

Number		
9(a)	$f'(x)=\lambda(x-2)(3x+1)$	M1
	$f(x)=\int\lambda(x-2)(3x+1)\,dx=\left[\int\lambda\left(3x^2-5x-2\right)\right]dx=\lambda\left(\frac{3x^3}{3}-\frac{5x^2}{2}-2x+k\right)$	M1A1
	At $(2,-9)$ $-9=2(2^3)-5(2^2)-4(2)+c\Rightarrow c=3$	M1
	$a=-5,b=-4,c=3\Rightarrow\left\{f(x)=2x^3-5x^2-4x+3\right\}$ *	A1*cso
		[5]
	ALT $f(x)=2x^3+ax^2+bx+c$ $f'(x)=6x^2+2ax+b$	[M1
	Both $\begin{cases}0=6(2^2)+2(2)a+b\\0=6\times\left(-\frac{1}{3}\right)^2+2\times\left(-\frac{1}{3}\right)\times a+b\end{cases}$	M1A1
	$\begin{cases}72+12a+3b=0 \text{ and } 12-12a+18b=0\Rightarrow a=...,b=...\\\text{At } (2,9)\quad -9=2(2^3)-5(2^2)-4(2)+c\Rightarrow c=...\end{cases}$	M1
	$a=-5,b=-4,c=3\Rightarrow\left\{f(x)=2x^3-5x^2-4x+3\right\}$ *	A1*cso]
(b)(i)	$f(-1)=2(-1)^3-5(-1)^2-4(-1)+3=0$ oe	B1 [1]
(b)(ii)	$x+1\overline{)2x^3-5x^2-4x+3}$ $\quad 2x^2-7x+3$	M1
	$2x^2-7x+3=(2x-1)(x-3)\Rightarrow(x+1)(x-3)(2x-1)$	M1A1 [3]
(c)	$A=\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx+\left\lvert\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx\right\rvert$	M1
	$A=\left[\frac{1}{2}x^4-\frac{5}{3}x^3-2x^2+3x\right]_{-1}^{\frac{1}{2}}+\left\lvert\left[\frac{1}{2}x^4-\frac{5}{3}x^3-2x^2+3x\right]_{\frac{1}{2}}^{3}\right\rvert$	M1
	$A=\left\{\left[\frac{1}{2}.\left(\frac{1}{2}\right)^4-\frac{5}{3}.\left(\frac{1}{2}\right)^3-2.\left(\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)\right]-\left[\frac{1}{2}.(-1)^4-\frac{5}{3}.(-1)^3-2.(-1)^2+3.(-1)\right]\right\}$ $+\left\lvert\left\{\left[\frac{1}{2}.(3)^4-\frac{5}{3}.(3)^3-2.(3)^2+3(3)\right]-\left[\frac{1}{2}.\left(\frac{1}{2}\right)^4-\frac{5}{3}.\left(\frac{1}{2}\right)^3-2.\left(\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)\right]\right\}\right\rvert$	M1
	$A=\left\{\frac{117}{32}+\left\lvert-\frac{1375}{96}\right\rvert\right\}=\frac{863}{48}$	A1
		[4]
		Total 13 marks

Type MS · Factorising and solving a cubic

Answer 2 of 22 · Question 9 part b.ii · 3 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For forming an expression for $f'(x)$ using the two turning points given. Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	M1	Attempts to integrate, see General Guidance for the definition of an attempt Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	A1	Correct integration including +c, Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	M1	$\lambda = 2$ is stated or implied, $\lambda k = c$ Substitution of $(2,-9)$ to their $f(x)$ (Does not have to be simplified) or substitution of $\left(-\frac{1}{3},\frac{100}{27}\right)$ (Does not have to be simplified)
	A1*cso	Obtains the **given** answer with no errors in the working
	ALT - starts with $f(x) = 2x^3 + ax^2 + bx + c$	
	M1	Attempts to differentiate, see General Guidance for the definition of an attempt
	M1	Equates differentiated expression to 0 and substitutes $x = 2$ **and** equates differentiated expression to 0 and substitutes $x = -\frac{1}{3}$
	A1	Two correct equations, simplified or unsimplified
	M1	Attempts to solve two **correct** equations simultaneously (including using calculators) to find a, b **and** substitution of their $a, b, (2,-9)$ to $f(x)$ to find c or substitution of their $a, b, \left(-\frac{1}{3},\frac{100}{27}\right)$ to $f(x)$ to find c
	A1*cso	Obtains the **given** answer with no errors in the working
(b)(i)	B1	Demonstrating that $x+1$ is a factor $f(-1) = 2(-1)^3 - 5(-1)^2 - 4(-1) + 3 = 0$ (factor theorem) If shown via polynomial division, the division must be fully correct and obtains 0 as the remainder.
(b)(ii)	M1	Dividing by $(x+1)$ to obtain a three-term quadratic quotient: $2x^2 - 7x + ...$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line : e.g. $(x+1)(x-3)(2x-1)$
	ALT I – equates coefficients	
	M1	For stating $2x^3 - 5x^2 - 4x + 3 = (x+1)(Ax^2 + Bx + C) \Rightarrow$ $2x^3 - 5x^2 - 4x + 3 = Ax^3 + (A+B)x^2 + (B+C)x + C \Rightarrow$ $A = 2, B = -7, C = ...$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line: e.g. $(x+1)(x-3)(2x-1)$
	ALT II - inspection	
	M1	For finding the quadratic factor. Minimum required is $(x+1)(2x^2 - 7x + C)$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line: e.g. $(x+1)(x-3)(2x-1)$

Type MS · Factorising and solving a cubic

Answer 2 of 22 · Question 9 part b.ii · 3 marks

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(c)	M1	For identifying the correct limits, uses a correct strategy to find the area above the x-axis and below the x-axis: $\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx+\left\lvert\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx\right\rvert$ or $\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx-\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx$ or implied by their working
	M1	For an attempt to integrate $2x^3-5x^2-4x+3$ See General Guidance for the definition of an attempt.
	M1	For clear substitution of **correct** limits into any **one** of their integrated expressions (must be a changed expression). Can be implied by **one** of the **correct exact** areas found from a **correctly** integrated expression
	A1	For correctly obtaining $\frac{863}{48}$ or $17\frac{47}{48}$

Type MS · Factorising and solving a cubic

Answer 2 of 22 · Question 9 part b.ii · 3 marks

2025 JUN · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 3 of 22 · Question 2 part c · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

2 (a)	$f(1)=1+p+q=-12$	M1
	$p+q=-13$	A1
	$f(4)=64+4p+q=30$	M1
	$4p+q=-34$	A1
	$3p=-21$	M1
	$p=-7$ and $q=-6$	A1 (6)
(b)	$3^3-7\times3-6=27-21-6=0$ *	B1 cso (1)
(c)	$(x-3)(x^2+3x+2)$	M1
	$(x-3)(x+2)(x+1)$	M1 A1 (3)
(d)	$x=3$ $x=-2$ $x=-1$	B1 ft (1)
		[11]

	Notes
(a)	
M1	For substitution of 1 into f(x)
A1	For $p+q=-13$ oe
M1	For substitution of 4 into f(x)
A1	For $4p+q=-34$ oe
M1	For solving simultaneously
A1	$p=-7$ and $q=-6$
(b)	
B1cso	For substituting 3 into f(x) and obtaining the given result
(c)	
M1	For $(x-3)(x^2+3x+2)$
M1	For factorising the quadratic
A1	$(x-3)(x+2)(x+1)$
(d)	
B1ft	For $x=3$ $x=-2$ $x=-1$ or follow through part (c)

Type MS · Factorising and solving a cubic

Answer 3 of 22 · Question 2 part c · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 4 of 22 · Question 4 part b · 3 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

Number		
4 (a)	$f(2)=2\times2^3+a\times2^2+b\times2+18=0$ $f'(x)=6x^2+2ax+b \Rightarrow f'(2)=6\times2^2+2\times a\times2+b=5$ $4a+2b+34=0$ $4a+b+19=0$ $\Rightarrow b=-15,\ a=-1$	M1 M1M1 A1 M1A1 [6]
(b)	$x-2\overline{)2x^3-x^2-15x+18}$ $2x^2+3x-9$ $2x^2+3x-9=(x+3)(2x-3)$ $\Rightarrow (x-2)(x+3)(2x-3)$	M1 M1A1 [3]
(c)	$x=2,\ -3,\ \frac{3}{2}$	B2ft [2]
		Total 11 marks

Type MS · Factorising and solving a cubic

Answer 4 of 22 · Question 4 part b · 3 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks
3(a)	$\frac{dy}{dx} = 2e^{2x}\left(x^2+1\right)+e^{2x}\left(2x\right)$	M1A1A1 [3]
(b)	When $x=0$ $\frac{dy}{dx} = 2\times1\times1+1\times0=2$ $\quad y=e^{2\times0}\left(0+1\right)=1$ $y-1=2\left(x-0\right)\Rightarrow y=2x+1$	B1B1 B1 [3]
		Total 6 marks
(a) **M1** **A1** **A1** **(b)** **B1** **B1** **B1**	Attempted use of the product rule. Sum of two terms (either way round) with $x^n \to x^{n-1}$ (Condone e^{2x} instead of $2e^{2x}$) Once the correct answer is seen ISW. This mark may be implied by the sum of two terms with one of the two terms correct. Either term correct Both terms correct When $x=0$ $\frac{dy}{dx}=2$ When $x=0$ $y=1$ $y=2x+1$	

Type MS · Factorising and solving a cubic

Answer 4 of 22 · Question 4 part b · 3 marks

2020 JAN · Paper 01 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks
4 (a)	$f(2)=2\times2^3+a\times2^2+b\times2+18=0$ $f'(x)=6x^2+2ax+b \Rightarrow f'(2)=6\times2^2+2\times a\times2+b=5$ $4a+2b+34=0$ $4a+b+19=0$ $\Rightarrow b=-15,\ a=-1$	M1 M1M1 A1 M1A1 [6]
(b)	$\begin{array}{r} 2x^2+3x-9 \\ x-2\overline{)2x^3-x^2-15x+18} \end{array}$ $2x^2+3x-9=(x+3)(2x-3)$ $\Rightarrow (x-2)(x+3)(2x-3)$	M1 M1A1 [3]
(c)	$x=2,\ -3,\ \frac{3}{2}$	B2ft [2]
		Total 11 marks
(a)		
M1	$f(2)=0$ leading to an equation in a and b	
M1	Attempt to differentiate	
M1	$f'(2)=5$ leading to an equation in a and b	
A1	$4a+2b+34=0$ and $4a+b+19=0$	
M1	Solving simultaneously	
A1	$b=-15,\ a=-1$	
(b)		
M1	Dividing by $x-2$ to obtain a 3TQ	
M1	Factorising the 3TQ	
A1	All 3 terms correct	
(c)		
B2 ft	$x=2,\ -3,\ \frac{3}{2}$ (**B1** for 2 correct)	

Type MS · Factorising and solving a cubic

Answer 4 of 22 · Question 4 part b · 3 marks

2020 JAN · Paper 01 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 5 of 22 · Question 5 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks
5(a)	$f(1)=2\times1^3+a\times1^2-14\times1+b=0 \quad\Rightarrow\quad a+b=12$ $f(4)=2\times4^3+a\times4^2-14\times4+b=39\Rightarrow16a+b=-33$ $\Rightarrow15a=-45\Rightarrow a=-3*$ $\Rightarrow b=15$	M1 M1 M1A1 cso B1 [5]
	ALT – polynomial division	
	$(2x^3+ax^2-14x+b)\div(x-1)=2x^2+(a+2)x-14+a+2$ and comparison of final step of division with b to obtain an equation **or** $(2x^3+ax^2-14x+b)\div(x-4)=2x^2+(a+8)x+(18+4a)$ r 39 and comparison of final step of division with obtaining remainder 39 to obtain an equation $(2x^3+ax^2-14x+b)\div(x-1)=2x^2+(a+2)x-14+a+2$ and comparison of final step of division with b to obtain an equation **and** $(2x^3+ax^2-14x+b)\div(x-4)=2x^2+(a+8)x+(18+4a)$ r 39 and comparison of final step of division with obtaining remainder 39 to obtain an equation $b=12-a$ $b+4(18+4a)=39$ $a=-3*$ $b=15$	[M1 M1 M1 A1cso B1]
(b)	$(x-1)\overline{)2x^3-3x^2-14x+15}$ $\quad 2x^2-x-15$ $f(x)=(x-1)(2x+5)(x-3)$	M1A1 dM1A1 [4]
(c)	$p=-\frac{5}{2}\quad q=1\quad r=3$	B1ft B1ft B1ft [3]

Type MS · Factorising and solving a cubic

Answer 5 of 22 · Question 5 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 5 · find part b in this mark scheme extract

(d)	$A = \left\lvert \int_1^3 \left(2x^3 - 3x^2 - 14x + 15\right) \mathrm{d}x \right\rvert + \int_{-\frac{5}{2}}^{1} \left(2x^3 - 3x^2 - 14x + 15\right) \mathrm{d}x$	M1
	$A = \left\lvert \left[\frac{2x^4}{4} - \frac{3x^3}{3} - \frac{14x^2}{2} + 15x\right]_1^3 \right\rvert + \left[\frac{2x^4}{4} - \frac{3x^3}{3} - \frac{14x^2}{2} + 15x\right]_{-\frac{5}{2}}^{1}$	M1*
	$A = \left\lvert \left[\left(\frac{3^4}{2} - 3^3 - 7\times 3^2 + 15\times 3\right) - \left(\frac{1^4}{2} - 1^3 - 7\times 1^2 + 15\times 1\right)\right] \right\rvert$ $+ \left[\left(\frac{1^4}{2} - 1^3 - 7\times 1^2 + 15\times 1\right) - \left(\frac{\left(-\frac{5}{2}\right)^4}{2} - \left(-\frac{5}{2}\right)^3 - 7\times\left(-\frac{5}{2}\right)^2 + 15\times\left(-\frac{5}{2}\right)\right)\right]$	dM1
	$A = 12 + \frac{1715}{32} = \frac{2099}{32}$ or 65.59375	A1 [4]
		Total 16 marks

Type MS · Factorising and solving a cubic

Answer 5 of 22 · Question 5 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 5 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For use of f (1) = 0 **or** use of f (4) = 39
	M1	For use of f (1) = 0 **and** use of f (4) = 39
	M1	For solving the simultaneous equations in a and b Must see at least one line of working before a or b obtained. Must see an equation in just one variable.
	A1cso	For showing $a = -3$* Must see expression only in terms of a
	B1	For the correct value of $b = 15$ seen. Note: this is an A mark in epen.
	ALT – polynomial division	
	M1	Correct method for forming one equation by polynomial division together with comparison of the final step of the division with the required result (no remainder for division by $(x-1)$ and remainder 39 for division by $(x-4)$)
	M1	Correct method for forming **both** equations by polynomial division together with comparison of the final step of the division with the required result (no remainder for division by $(x-1)$ and remainder 39 for division by $(x-4)$)
	M1	For solving the simultaneous equations in a and b Must see at least one line of working before a or b obtained.
	A1cso	For showing $a = -3$* Must see expression only in terms of a ie $ka = -3k$ before $a = -3$
	B1	For the correct value of $b = 15$ seen. Note: this is an A mark in epen.
	Assuming $a = -3$ and solving to find $b = 15$ scores maximum M1M0M0A0A1	
(b)	**Condone (b) completed in (c)**	
	M1	For carrying out polynomial division/comparing coefficients to find the quadratic factor. Must reach $2x^2 \pm \cdots$. Allow statement of quadratic factor if polynomial division by $(x-1)$ used in (a).
	A1	For the correct quadratic factor.
	dM1	For factorising their 3TQ See general guidance for what constitutes an attempt to factorise a 3TQ.
	A1	For the correct factorisation of f (x) on one line. Condone omission of f$(x) =$
	Note: complete factorisation seen with no method scores M0A0M0A0.	
(c)	B1ft	For one correct value for any of p, r or q FT their linear factors in (b). Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.
	B1ft	For two correct values. FT their linear factors in (b). Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.
	B1ft	For all three correct values. FT their linear factors in (b), but **1 must be one of the values**. Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.

Type MS · Factorising and solving a cubic

Answer 5 of 22 · Question 5 part b · 4 marks

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(d)	M1	For a correct statement for the area of the shaded regions with their limits correctly used. Minimum required is: $\int_p^q (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x - \int_q^r (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x$ with $p < q < r$ shown as numerical limits or $\int_p^q (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x + \left\lvert \int_q^r (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x \right\rvert$ with $p < q < r$ shown as numerical limits
	M1*	For an attempt to integrate their expression. Minimum required is: • two powers of terms to increase by 1 • no powers of terms to decrease Condone $+c$
	dM1	For substituting their limits correctly to evaluate the integrated expression. Must show the substitution if their integral or their limits are incorrect. Correct exact answer following correct integral with correct limits implies this mark. Condone $+c$ as long as this is cancelled out during the substitution.
	A1	For the correct area. Exact area required.

Type MS · Factorising and solving a cubic

Answer 5 of 22 · Question 5 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 5 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 6 of 22 · Question 4 part b · 3 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

4(a)	$f'(x) = 3px^2 + 2qx - 37$ $f'(-2) = 3p(-2)^2 + 2q(-2) - 37 = -33$ $f(-5) = p(-5)^3 + q(-5)^2 - 37(-5) - 12q = 0$ $\begin{cases} 12p - 4q = 4 \\ -125p + 13q = -185 \end{cases} \Rightarrow p = \cdots \; q = \cdots$ $p = 2, q = 5$	M1 M1 M1 dM1 A1A1 [6]
(b)	$x+5\overline{)2x^3+"5"x^2-37x-"60"}$ $\quad 2x^2-"5"x-"12"$ OR $2x^3 + "5"x^2 - 37x - "60" = (x+5)(Ax^2 + Bx + C)$ $\Rightarrow A = 2, B = \cdots, C = \cdots$ $2x^2 - 5x - 12 = (2x+3)(x-4)$ $\{f(x) =\}(x+5)(2x+3)(x-4)$	M1 M1 A1 [3]
(c)	$(x+5)(2x+3)(x-4) = 0, \Rightarrow x = -5,\ 4,\ -\frac{3}{2}$	M1A1 [2]
		Total 11 marks

Part	Mark	Notes
(a) Mark i and ii together	**M1**	For differentiating the given expression in terms of p and q with respect to x, at least one term correct and no terms integrated.
	M1	Attempts $f'(\pm 2) = -33$
	M1	Attempts $f(\pm 5) = 0$
	dM1	For solving the two resulting simultaneous equations by any correct method (allow slips) providing there are two variables p and q in both, leading a value of p or q. Explicit method to solve must been seen. **Depends on all previous method marks**
	A1	For $p = 2$ or $q = 5$ from correct working in two correct equations
	A1	For both $p = 2$ and $q = 5$ from correct working
(b)	**M1**	For using polynomial division (or any other **algebraic** method) to find the quadratic factor. **Division:** Look for divisor of $(x+5)$, quotient of a quadratic $2x^2 -$ their $qx + c$ $(c \neq 0)$ **Compare coefficients**: minimum required is an identity of the correct form followed by values of $A = 2, B = -$their $q, C = \ldots (C \neq 0)$
	M1	Correct method of factorising their quadratic factor. (See General Principles)
	A1	Correct factorised expression written in full on one line.
(c)	**M1**	Correct method of solving their $f(x) = 0$. Their $f(x)$ must have 3 solutions, one of which must be $x = -5$, "=0" or their method may be implied by their x values. They must use their result in part b which $f(x)$ reaches 2 or 3 factors, if part b is not attempted, a full algebraic method of solving $f(x)=0$ must be seen in this part.
	A1	For all three correct values of x, no extra solutions.

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(a) **ALT I Algebraic Division**

M1 Same as main scheme
M1 Their f′(*x*) is in the form of a 3TQ, look for divisor of (*x*+2), quotient of a linear expression leading to a constant remainder in ***p*** and ***q***, sets = -33
M1 For dividing f(*x*), look for divisor of (*x*+5), quotient of a 3TQ, leading to a constant remainder in ***p*** and ***q***, sets=0
dM1A1 Same as main scheme

You may also see variations for their division, marking is similar as above, (if not sure, send to review)
e.g.
For f′(*x*)

$$\begin{array}{r|lll} -2 & 3p & 2q & -37 \\ & & -6p & 12p-4q \\ \hline & 3p & -7p+2q & 12p-4q-37=-33 \end{array}$$

For f(*x*)

$$\begin{array}{r|llll} -5 & p & q & -37 & -12q \\ & & -5p & -5q+25p & 25q-125p+185 \\ \hline & p & q-5p & -5+25p-37 & 13q-125p+185=0 \end{array}$$

(a) **ALT II Compare coefficients**
M1 Same as main scheme
M1 Sets their f′(*x*) which is in the form of a 3TQ, equal to (*x*+2)(K*x*+C)-33 ($K \neq 0, C \neq 0$) , then attempts to equate the coefficients of *x* to form an equation in ***p*** and ***q***
M1 Sets f(*x*)=(*x*+5) [$px^2 + Rx + S$] ($R \neq 0, S \neq 0$), then attempts to equate their coefficients of *x* to form an equation in ***p*** and ***q***.
dM1A1 Same as main scheme

For reference, the correct working for this method:

$$\begin{aligned} f'(x) &= 3px^2 + 2qx - 37 \\ &= (x+2)(3px-2) - 33 \\ \Rightarrow 6px - 2x &= 2qx \\ 6p - 2 &= 2q \end{aligned}$$

$$f(x) = px^3 + qx^2 - 37x - 12q = (x+5)\left[px^2 + (q-5p)x - \frac{12q}{5}\right]$$

$$\begin{aligned} \Rightarrow 5(q-5p)x - \frac{12q}{5}x &= -37x \\ 5(q-5p) - \frac{12q}{5} &= -37 \\ \frac{13q}{5} - 25p &= -37 \end{aligned}$$

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Question	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 7 of 22 · Question 6 part d · 2 marks

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Number		
6 (a)	$f(3) = 27 + 9p + 9 - 30 + q = 0$	M1 A1
	$9p + q + 6 = 0$ *	A1 cso
		(3)
(b)	$f(-p) = -p^3 + p^2(p+1) + 10p + q = 0$	M1 A1
	$p^2 + 10p + q = 0$ *	A1 cso
		(3)
(c)	$p^2 + 10p - 9p - 6 = 0$	M1
	$p^2 + p - 6 = 0$	A1
	$(p+3)(p-2) = 0$	M1
	$p = 2 \quad q = -24$	A1 A1
		(5)
(d)	$(x+a)(x-3)(x+2)$	
	So $-3 \times 2 \times a = -24 \quad a = 4$	M1
	$(x+4)(x-3)(x+2)$	A1
		(2)
		Total 13 marks

Part	Mark	Guidance
General guidance for marking parts (a) and (b) • For the award of full marks in parts (a) and/or (b) you **must** see = 0 used in a line of working **before** the final answer. • If a candidate does not use = 0 in either parts (a) or (b) [except in the final line – which is a given answer] deduct the M mark (and the subsequent A marks) in only the **first** occurrence of the absence.		
(a)	**M1**	For using $f(\pm 3) = 0$ in the given equation set = 0
	A1	For obtaining the correct unsimplified expression: $27 + 9p + 9 - 30 + q = 0$
	A1 cso	For obtaining the given equation $9p + q + 6 = 0$ * **Note: This is a show question. There must be no errors seen.**
(b)	**M1**	For use of $f(\pm p) = 0$ in the given equation set = 0
	A1	For obtaining the correct unsimplified expression: $-p^3 + p^2(p+1) + 10p + q = 0$

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	A1 cso	For obtaining the correct given equation $p^2+10p+q=0$ * **Note: This is a show question. There must be no errors seen.**
(c)	**M1**	For attempting to solve the given two equations simultaneously to achieve a 3TQ in either p or q only. E.g. substitutes $q=\mp 9p\mp 6$ or $\left[p=\frac{\mp q\mp 6}{9} \text{ and } p^2=\frac{(\mp q\mp 6)^2}{81}\right]$ into $p^2+10p+q=0$ This mark may be implied by the correct 3TQ
	A1	For the correct 3TQ $p^2+p-6=0$ or $q^2+3q-504=0$
	M1	For an attempt to solve their 3TQ in either p or q using factorisation, use of the formula or completing the square. See general guidance for the definition of an attempt. For example: $(p+3)(p-2)=0\Rightarrow p=\ldots$ or $(q+24)(q-21)=0\Rightarrow q=\ldots,\ \ldots$ If a candidate uses their calculator to solve their 3TQ, the final values must be correct for the award of this mark unless a valid method is seen.
	A1	For either the correct value of p OR the correct value of q $p=2$ or $q=-24$ Condone the presence $p=-3$, and/or $q=21$
	A1	For both the correct value of p (= 2) AND the correct value of q (= –24) Must reject $p=-3$, and/or $q=21$ if seen.
(d)	**M1**	[f(x) = $(x+a)(x-3)(x+'2')$] For attempting to find the value of a $-3\times'2'\times a='-24'\Rightarrow a=\ldots$ OR For an attempt using division with their values of p and q $x^2-x-6\overline{)x^3+3x^2-10x-24}$ $\quad x+4$ or $x-3\overline{)x^3+3x^2-10x-24}$ $\quad x^2+6x+8$ Allow a quotient of $x+b$ or x^2+6x+b where b is a constant.
	A1	For the correct factorised expression $(x+4)(x-3)(x+2)$ which must be written out in full on one line.

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Question number	Scheme	Marks

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Answer 8 of 22 · Question 9 part a · 4 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

9 a	$(ax^4 = 3x^4) \Rightarrow a = 3$	B1
	$(4c = 64) \Rightarrow c = 16$	B1
	$(bx^3 - 4ax^3 = 4x^3) \Rightarrow b - 4a = 4$ or $(4ax^2 - 4bx^2 + cx^2 = -36x^2)$ $\Rightarrow 4a - 4b + c = -36$ *or* $(4bx - 64x = 0x) \Rightarrow 4b - 64 = 0$	M1
	$b = 16$	A1 (4)
b	$\int_0^2 (x^3 + x^2 - 6x)\mathrm{d}x = \left[\frac{x^4}{4} + \frac{x^3}{3} - \frac{6}{2}x^2\right]_0^2$	B1 (M1 on ePen) M1
	$= \left("\frac{2^4}{4} + \frac{2^3}{3} - 3(2)^2"\right) - (0)$	M1 (A1 on ePen)
	$= \pm\frac{16}{3}$ oe	A1 (M1 on ePen)
	$\left(\left["\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right]_x^0 = \pm\frac{16}{3} \Rightarrow\right) (0) - \left("\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right) = \pm\frac{16}{3}$ oe	M1
	$3x^4 + 4x^3 - 36x^2 + 64 = 0$	A1
	$(x-2)^2(3x^2 + 16x + 16) = 0$ *	A1* cso (7)
ALT	$\int_x^2 (x^3 + x^2 - 6x)\mathrm{d}x = \left["\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right]_x^2$	B1 (M1 on ePen) M1
	$= \left("\frac{2^4}{4} + \frac{2^3}{3} - 3(2)^2"\right) - \left("\frac{x^4}{4} + \frac{x^3}{3} - 3(x)^2"\right)$	M1 (A1 on ePen)
	$\pm\left(-\frac{16}{3} - \frac{x^4}{4} - \frac{x^3}{3} + 3x^2\right) = 0$	A1 (M1 on ePen)
	$3x^4 + 4x^3 - 36x^2 + 64 = 0$	M1 A1
	$(x-2)^2(3x^2 + 16x + 16) = 0$ *	A1
c	(When $(x-2) = 0$ $x = 2$ and when $3x^2 + 16x + 16 = 0$) $(3x+4)(x+4) = 0$ so $x = -\frac{4}{3}$ and $x = -4$	M1 A1
	($x \neq -4$ as it is to the left of the point at $x = -3$) $x = -\frac{4}{3}$	M1
	When "$x = -\frac{4}{3}$" $y = \left("-\frac{4}{3}"\right)^3 + \left("-\frac{4}{3}"\right)^2 - 6\left("-\frac{4}{3}"\right) = \frac{200}{27}$	dM1
	So $A = \left(-\frac{4}{3}, \frac{200}{27}\right)$	A1 (5)
		Total 16 marks

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Part	Mark	Notes
(a)	B1	$a = 3$
	B1	$c = 16$
	M1	For a clear process to equate coefficients to find either of the equations $b - 4a = 4$ or $4a - 4b + c = -36$ or $4b - 64 = 0$. Allow one error. For algebraic division: there must be a complete attempt to divide by $x^2 + px + q$ and reach an expression of the form $3x^2 + rx + 16,\ p, q, r \neq 0$
	A1	$b = 16$
Full marks may be awarded for all of *a, b* and *c* appearing correctly with no method shown. Embedded *a, b* and *c* should be awarded full marks.		
(b)	B1 (M1 on ePen)	Correct limits of $x = 0$ and $x = 2$ – or used later in working.
	M1	For a minimally acceptable attempt to integrate any 3 term cubic of the form $x^3 + fx^2 + gx$. Limits do not need to be present. See general guidance for the definition of a minimally acceptable attempt.
	M1 (A1 on ePen)	Substitution of their limits into any changed expression, the correct way round minimum 3 terms. Sub of 0 not needed. Allow this mark to be implied by a fully correct value.
	A1 (M1 on ePen)	Either for a fully correct substitution, or for $\pm\frac{16}{3}$
	M1	For $(0) - \left("\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right) = \pm\frac{16}{3}$ oe. Allow use of their integrated expression and their $\pm\frac{16}{3}$ which can be in any equivalent form, including an unevaluated expression. Allow use of *n* rather than *x*. The limit of 0 need not be seen.
	A1	For $3x^4 + 4x^3 - 36x^2 + 64 = 0$ Allow use of *n* rather than *x*. $\frac{16}{3}$ must have been evaluated.
	A1* cso	For $(x-2)^2(3x^2 + 16x + 16) = 0$ .No errors in working. Candidates can use *n* throughout.
ALT	B1 (M1 on ePen)	Correct limits of $x = x$ and $x = 2$ – seen on the integral or used later in working. These can be either way round.
	M1	For a minimally acceptable attempt to integrate any 3 term cubic of the form $x^3 + fx^2 + gx$. Limits do not need to be present. See general guidance for the definition of a minimally acceptable attempt.
	M1 (A1 on ePen)	For substitution of both of their limits into any changed expression. Minimum 3 terms
	A1 (M1 on ePen)	For $\pm\left(-\frac{16}{3} - \frac{x^4}{4} - \frac{x^3}{3} + 3x^2\right) = 0$
	M1	For a valid attempt to multiply their equation throughout to arrive at integer coefficients.
	A1 A1	As main scheme. Candidates can use *n* throughout.
(c)	M1	For a complete and minimally acceptable attempt to solve $3x^2 + 16x + 16 = 0$ to give $x =$ See general guidance for definition of minimally acceptable. Some candidates may have factorised in part b), if used in part c), marks can be awarded
	A1	For $x = -\frac{4}{3}$ and $x = -4$
	M1	Chooses $"x = -\frac{4}{3}"$, can be stated explicitly or used later implicitly. Allow choice of their *x* coordinate for $-3 < x < 0$
	dM1	For substituting $x = "-\frac{4}{3}"$ into $y = x(x+3)(x-2)$ Allow substitution of their *x*, dependent on previous method mark
	A1	$A = \left(-\frac{4}{3}, \frac{200}{27}\right)$ students may list $x = -\frac{4}{3}, y = \frac{200}{27}$

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Question number	Scheme	Marks
	→ → 3 → 3 → 1 → 1	

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Question	Scheme	Marks
5(a)	$f(-2)=-16a+4+2b+3a=0$ and $f(1)=2a+1-b+3a=0$	M1M1
	$-13a+4+2b=0$ and $5a+1-b=0 \Rightarrow a=2^*$ $b=11$	M1A1cso A1 [5]
(b)	$f(x)=4x^3+x^2-11x+6=(4x-3)(x+2)(x-1)$	M1A1 [2]
(c)	$h(y)=2^{(3y+2)}+2^{2y}-11(2^y)+6 \Rightarrow h(y)=4(2^y)^3+(2^y)^2-11(2^y)+6$	M1
	$x=2^y \Rightarrow 2^y-1=0,\ \ 2^y+2=0,\ \ 4(2^y)-3=0$	M1
	$2^y=1 \Rightarrow y=0$	B1
	$2^y=\frac{3}{4} \Rightarrow y=\log_2\frac{3}{4}=-0.4150... \Rightarrow y=-0.415$ $[2^y=-2$ no solution$]$	M1A1 [5]
		Total 12 marks

Question	Notes	Marks
(a)	For substituting ± 2 into $f(x)=0$ Allow sign errors	M1
	For substituting ± 1 into $f(x)=0$ Allow sign errors	M1
	For solving the resulting pair of simultaneous equations. $-13a+4+2b=0$ $5a+1-b=0$	M1
	For $a=2$ *	A1 cso
	For $b=11$	A1 [5]
	ALT uses polynomial division.	
	Multiplies out the product of the two factors to give: $(x+2)(x-1)=x^2+x-2$ This must be correct.	M1

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	Uses polynomial division to give as a minimum: $2ax+(1-2M)$ where M is a constant. $x^2+x-2\overline{)2ax^3+x^2-bx+3a}$ $\quad 2ax+(1-2a)$	M1
	Equates coefficients: $(x^2+x-2)(2ax+(1-2a))=2ax^3+x^2-bx+3a$ $\Rightarrow 2ax^3+x^2+x(1-6a)+(-2+4a)=2ax^3+x^2-bx+3a$ $3a=4a-2$ $1-6a=-b$	M1
	For $a=2$ *	A1 cso
	For $b=11$	A1
(b)	Divides $f(x)=4x^3+x^2-11x+6$ by either $[x^2+x-2]$ or $x+2$ or $x-1$ Either, $\left[x^2+x-2\overline{)4x^3+x^2-11x+6}\right.$ $\quad 4x+k$ $x+2\overline{)4x^3+x^2-11x+6}$ $\quad 4x^2-7x\pm k$ $x-1\overline{)4x^3+x^2-11x+6}$ $\quad 4x^2+5x\pm k$	M1
	For the correct factorisation of $f(x)$ $4x^2-7x+3=(4x-3)(x-1)\Rightarrow f(x)=(4x-3)(x-1)(x+2)$	A1 [2]
(c)	For manipulating the indices to achieve $\Rightarrow h(y)=4(2^y)^3+(2^y)^2-11(2^y)+6$ OR Uses the factorised expression to obtain: $h(y)=(2^y+2)(2^y-1)(4(2^y)-3)$	M1
	Substitutes $x=2^y$ into the factorised f (x) to find either $2^y-1=0$ or $2^y+2=0$ or $4(2^y)-3=0$	M1
	For $2^y=1\Rightarrow y=0$	B1
	For $4(2^y)=3\Rightarrow 2^y=\frac{3}{4}\Rightarrow y=\log_2\frac{3}{4}$	M1
	For awrt $y=-0.4150...\Rightarrow y\approx -0.415$ [No solution for $2^y=-2$ - must reject]	A1 [5]
		Total 12 marks

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Question	Scheme	Marks

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Question	Scheme	Marks
5(a)	$f(1)=2\times1^3+a\times1^2-14\times1+b=0 \quad \Rightarrow \quad a+b=12$ $f(4)=2\times4^3+a\times4^2-14\times4+b=39 \Rightarrow 16a+b=-33$ $\Rightarrow 15a=-45 \Rightarrow a=-3^*$ $\Rightarrow b=15$	M1 M1 M1A1 cso B1 [5]
	ALT – polynomial division	
	$(2x^3+ax^2-14x+b)\div(x-1)=2x^2+(a+2)x-14+a+2$ and comparison of final step of division with b to obtain an equation **or** $(2x^3+ax^2-14x+b)\div(x-4)=2x^2+(a+8)x+(18+4a)$ r 39 and comparison of final step of division with obtaining remainder 39 to obtain an equation $(2x^3+ax^2-14x+b)\div(x-1)=2x^2+(a+2)x-14+a+2$ and comparison of final step of division with b to obtain an equation **and** $(2x^3+ax^2-14x+b)\div(x-4)=2x^2+(a+8)x+(18+4a)$ r 39 and comparison of final step of division with obtaining remainder 39 to obtain an equation $b=12-a$ $b+4(18+4a)=39$ $a=-3^*$ $b=15$	[M1 M1 M1 A1cso B1]
(b)	$(x-1)\overline{)2x^3-3x^2-14x+15}$ $\quad 2x^2-x-15$ $f(x)=(x-1)(2x+5)(x-3)$	M1A1 dM1A1 [4]
(c)	$p=-\frac{5}{2} \quad q=1 \quad r=3$	B1ft B1ft B1ft [3]

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(d)	$A = \left\lvert \int_1^3 \left(2x^3 - 3x^2 - 14x + 15\right) \mathrm{d}x \right\rvert + \int_{-\frac{5}{2}}^{1} \left(2x^3 - 3x^2 - 14x + 15\right) \mathrm{d}x$	M1
	$A = \left\lvert \left[\frac{2x^4}{4} - \frac{3x^3}{3} - \frac{14x^2}{2} + 15x\right]_1^3 \right\rvert + \left[\frac{2x^4}{4} - \frac{3x^3}{3} - \frac{14x^2}{2} + 15x\right]_{-\frac{5}{2}}^{1}$	M1*
	$A = \left\lvert \left[\left(\frac{3^4}{2} - 3^3 - 7\times 3^2 + 15\times 3\right) - \left(\frac{1^4}{2} - 1^3 - 7\times 1^2 + 15\times 1\right)\right] \right.$ $\left. + \left[\left(\frac{1^4}{2} - 1^3 - 7\times 1^2 + 15\times 1\right) - \left(\frac{\left(-\frac{5}{2}\right)^4}{2} - \left(-\frac{5}{2}\right)^3 - 7\times\left(-\frac{5}{2}\right)^2 + 15\times\left(-\frac{5}{2}\right)\right)\right] \right\rvert$	dM1
	$A = 12 + \frac{1715}{32} = \frac{2099}{32}$ or 65.59375	A1 [4]
		Total 16 marks

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Part	Mark	Notes
(a)	M1	For use of f (1) = 0 **or** use of f (4) = 39
	M1	For use of f (1) = 0 **and** use of f (4) = 39
	M1	For solving the simultaneous equations in a and b Must see at least one line of working before a or b obtained. Must see an equation in just one variable.
	A1cso	For showing $a = -3$* Must see expression only in terms of a
	B1	For the correct value of $b = 15$ seen. Note: this is an A mark in epen.
	ALT – polynomial division	
	M1	Correct method for forming one equation by polynomial division together with comparison of the final step of the division with the required result (no remainder for division by $(x-1)$ and remainder 39 for division by $(x-4)$)
	M1	Correct method for forming **both** equations by polynomial division together with comparison of the final step of the division with the required result (no remainder for division by $(x-1)$ and remainder 39 for division by $(x-4)$)
	M1	For solving the simultaneous equations in a and b Must see at least one line of working before a or b obtained.
	A1cso	For showing $a = -3$* Must see expression only in terms of a ie $ka = -3k$ before $a = -3$
	B1	For the correct value of $b = 15$ seen. Note: this is an A mark in epen.
	Assuming $a = -3$ and solving to find $b = 15$ scores maximum M1M0M0A0A1	
(b)	**Condone (b) completed in (c)**	
	M1	For carrying out polynomial division/comparing coefficients to find the quadratic factor. Must reach $2x^2 \pm \cdots$. Allow statement of quadratic factor if polynomial division by $(x-1)$ used in (a).
	A1	For the correct quadratic factor.
	dM1	For factorising their 3TQ See general guidance for what constitutes an attempt to factorise a 3TQ.
	A1	For the correct factorisation of f (x) on one line. Condone omission of f$(x) =$
	Note: complete factorisation seen with no method scores M0A0M0A0.	
(c)	B1ft	For one correct value for any of p, r or q FT their linear factors in (b). Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.
	B1ft	For two correct values. FT their linear factors in (b). Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.
	B1ft	For all three correct values. FT their linear factors in (b), but **1 must be one of the values**. Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.

Type MS · Factorising and solving a cubic

Answer 10 of 22 · Question 5 part c · 3 marks

2023 JUN · Paper 01 (regional R) · extract 4 of 5 · find part c in this mark scheme extract

(d)	M1	For a correct statement for the area of the shaded regions with their limits correctly used. Minimum required is: $\int_p^q (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x - \int_q^r (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x$ with $p < q < r$ shown as numerical limits or $\int_p^q (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x + \left\lvert \int_q^r (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x \right\rvert$ with $p < q < r$ shown as numerical limits
	M1*	For an attempt to integrate their expression. Minimum required is: • two powers of terms to increase by 1 • no powers of terms to decrease Condone $+c$
	dM1	For substituting their limits correctly to evaluate the integrated expression. Must show the substitution if their integral or their limits are incorrect. Correct exact answer following correct integral with correct limits implies this mark. Condone $+c$ as long as this is cancelled out during the substitution.
	A1	For the correct area. Exact area required.

Type MS · Factorising and solving a cubic

Answer 10 of 22 · Question 5 part c · 3 marks

2023 JUN · Paper 01 (regional R) · extract 5 of 5 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 11 of 22 · Question 5 part b · 4 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

5(a)	$f(-2) = 0, \quad f(-3) = 21$ $a(-2)^3 + 5b(-2)^2 + 8a(-2) - 4b = 0$ $a(-3)^3 + 5b(-3)^2 + 8a(-3) - 4b = 21$ $2b = 3a$ $41b = 51a + 21$ $a = 2^*, b = 3$	M1 A1 M1 A1cso A1 [5]
(b)	$x+2\overline{)2x^3+15x^2+16x-12}$ $\quad 2x^2+11x-6$ **ALT** $2x^3+15x^2+16x-12=(x+2)(Ax^2+Bx+C)\Rightarrow(x+2)(2x^2+11x-6)$ $2x^2+11x-6=(2x-1)(x+6)$ $(2x-1)(x+6)=0 \Rightarrow 2x-1=0, x+6=0$ $x=-6, \ -2, \ \frac{1}{2}$	M1 M1 M1 A1 [4]
		Total 9 marks

Type MS · Factorising and solving a cubic

Answer 11 of 22 · Question 5 part b · 4 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

		Equates coefficients to find the 3TQ factor $2x^3+15x^2+16x-12=(x+2)(Ax^2+Bx+C)\Rightarrow(x+2)(2x^2+11x-6)$ Must get as far as $(x+2)(2x^2+\cdots)$ for the mark.
	M1	For attempting to factorise **their** 3TQ, but it must be a 3TQ $2x^2+11x-6=(2x-1)(x+6)$ Refer to general guidance for what constitutes an attempt to factorise.
	M1	An attempt to solve f $(x) = 0$
	A1	For $x=-6,\ -2,\ \frac{1}{2}$ **Note:** Correct answers with no working scores M0M0M0A0

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Factorising and solving a cubic

Answer 11 of 22 · Question 5 part b · 4 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

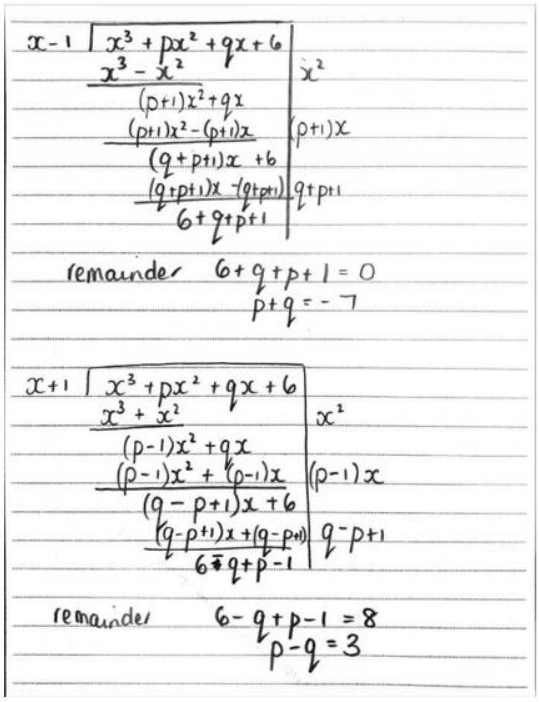
Question number	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 12 of 22 · Question 4 part b · 3 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

4 a Mark parts (i) and (ii) together	$(f(1)=)1^3+p\times1^2+q\times1+6=0$ oe	M1
	or	
	$(f(-1)=)(-1)^3+p\times(-1)^2+q\times(-1)+6=8$ oe	
	$p+q=-7$ oe	A1
	$p-q=3$ oe	A1
	Solving simultaneously $2p=-4$ or $2q=-10$	M1
	$p=-2$ *	A1 cso
	and $q=-5$	B1 (A1 on ePen) cso
		(6)
b	$(f(x)=x^3-2x^2-5x+6=0)$	
	$(x-1)(x^2-x-6)=0$	M1
	$[(x-1)](x+2)(x-3)=0$	M1
	$x=1,-2,3$	A1
		(3)
		Total 9 marks



Type MS · Factorising and solving a cubic

Answer 12 of 22 · Question 4 part b · 3 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For a fully correct substitution of either 1 into f(x) = 0 **OR** -1 into f(x) = 8 as shown. The bracketing for the substitution of -1 must be correct, but may be recovered later. A correct equation can imply this mark.
	A1	For a fully correct substitution of 1 into f(x) = 0 **AND** -1 into f(x) = 8 as shown. The bracketing for the substitution of -1 must be correct, but may be recovered later. A correct equation can imply this mark.
	A1	For $p+q=-7$ **and** $p-q=3$ oe The equations do not need to be simplified.
	Note, if students choose to do long division, to be comparable with the main scheme, this must be fully correct and complete, to allow them to arrive at the correct equations to be awarded the method mark. See the example included. The accuracy marks can be then awarded as stated above. There is an example under the main mark scheme. **If in any doubt, please send to review.**	
	M1	For an attempt to solve their equations simultaneously, allow one processing error. Must eliminate one variable and arrive at a value for p or q.
	A1 cso	For $p=-2$
	B1 (A1 on ePen)	$q=-5$ Note this is an independent accuracy mark, allowing students to use the given value of p to find q.
(b)	M1	For $(x-1)(x^2 \pm Ax-6)=0 \qquad A\neq 0$ This mark can be awarded for sight of $x^2 \pm Ax-6 \qquad A\neq 0$
	M1	For $[(x-1)](x+2)(x-3)=0$ **For this mark, for the correct or ft their quadratic:** Allow any minimally acceptable attempt to solve the quadratic, by factorising, completing the square or use of the quadratic formula – see general guidance, leading to two values of x in addition to x = 1. It is not necessary to see = 0.
	A1	For $x=1,-2,3$ If 3 three correct solutions appear with no working M1 M1 A1 as the question has not stated show working or solve algebraically.

Type MS · Factorising and solving a cubic

Answer 12 of 22 · Question 4 part b · 3 marks

2023 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
	$\left(\quad k\quad\right)\quad\quad\left(\quad k\quad\right)\quad\quad kx^2+(k+1)x-1\quad k$	

Type MS · Factorising and solving a cubic

Answer 13 of 22 · Question 2 part d · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

2 (a)	$f(1)=1+p+q=-12$	M1
	$p+q=-13$	A1
	$f(4)=64+4p+q=30$	M1
	$4p+q=-34$	A1
	$3p=-21$	M1
	$p=-7$ and $q=-6$	A1 (6)
(b)	$3^3-7\times3-6=27-21-6=0$ *	B1 cso (1)
(c)	$(x-3)(x^2+3x+2)$	M1
	$(x-3)(x+2)(x+1)$	M1 A1 (3)
(d)	$x=3$ $x=-2$ $x=-1$	B1 ft (1)
		[11]

	Notes
(a)	
M1	For substitution of 1 into f(x)
A1	For $p+q=-13$ oe
M1	For substitution of 4 into f(x)
A1	For $4p+q=-34$ oe
M1	For solving simultaneously
A1	$p=-7$ and $q=-6$
(b)	
B1cso	For substituting 3 into f(x) and obtaining the given result
(c)	
M1	For $(x-3)(x^2+3x+2)$
M1	For factorising the quadratic
A1	$(x-3)(x+2)(x+1)$
(d)	
B1ft	For $x=3$ $x=-2$ $x=-1$ or follow through part (c)

Type MS · Factorising and solving a cubic

Answer 13 of 22 · Question 2 part d · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 14 of 22 · Question 10 part b · 4 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

10 (a)	$f(1) = 32(1^3) - 33(1) + 1 = 0$ *	B1 cso (1)
(b)	$(x-1)(32x^2+32x-1)=0$ A correct method shown to solve a quadratic	M1
	e.g. $\frac{-32 \pm \sqrt{32^2+4\times32}}{64}$	M1
	$(x=1)$ or $\frac{-4+3\sqrt{2}}{8}$ or $\frac{-4-3\sqrt{2}}{8}$ Accept decimals correct to 3 sf e.g. = 0.0303… or = –1.03 …	A1 A1 (4)
(c)	$\sqrt{x} = \frac{1}{8x}$	M1
	$p = \frac{1}{4}$	A1 (2)

	Notes
(a)	
B1 cso	For substitution of 1 into $f(x)$ to obtain the given result
(b)	
M1	For $(x-1)(32x^2+32x-1)=0$
M1	A correct method shown to solve a quadratic. If an algebraic method is not shown then M0A0A0 is awarded.
A1	For $\frac{-4+3\sqrt{2}}{8}$ Allow 0.0303 or better
A1	For $\frac{-4-3\sqrt{2}}{8}$ Allow -1.03 or better
(c)	
M1	For equating the two equations
A1	For $p = \frac{1}{4}$

Type MS · Factorising and solving a cubic

Answer 14 of 22 · Question 10 part b · 4 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

(d)	$\pi\int_{\frac{1}{4}}^{a} x\,\mathrm{d}x = \pi\left[\frac{x^2}{2}\right]_{\frac{1}{4}}^{a}$	M1
	$= \pi\left(\frac{a^2}{2} - \frac{1}{32}\right)$	A1
	$\pi\int_{\frac{1}{4}}^{a} \frac{1}{64x^2}\,\mathrm{d}x = \pi\left[-\frac{1}{64x}\right]_{\frac{1}{4}}^{a}$	M1
	$= \pi\left(\frac{1}{16} - \frac{1}{64a}\right)$	A1
	$= \pi\left(\frac{a^2}{2} - \frac{1}{32}\right) - \pi\left(\frac{1}{16} - \frac{1}{64a}\right) = \frac{27\pi}{64}$	dM1
	$32a^3 - 33a + 1 = 0$	A1
	So $a = 1$	A1 (7)
	Alternative Method	
	$\pi\int_{\frac{1}{4}}^{a}\left(x - \frac{1}{64x^2}\right)\mathrm{d}x = \frac{27\pi}{64}$	M1
	$\int_{\frac{1}{4}}^{a}\left(64x - \frac{1}{x^2}\right)\mathrm{d}x = 27$	A1
	$\left[32x^2 + \frac{1}{x}\right]_{\frac{1}{4}}^{a} = 27$	M1 A1
	$\left(32a^2 + \frac{1}{a}\right) - (2+4) = 27$	dM1
	$32a^3 - 33a + 1 = 0$	A1
	So $a = 1$	A1 (7) **[14]**

Type MS · Factorising and solving a cubic

Answer 14 of 22 · Question 10 part b · 4 marks

2020 NOV · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

	Notes
(d)	
M1	For an attempt to integrate $\pi\int_{\frac{1}{4}}^{a} x\ \mathrm{d}x$ Ignore limits
A1	For $\pi\left(\frac{a^2}{2}-\frac{1}{32}\right)$
M1	For an attempt to integrate $\pi\int_{\frac{1}{4}}^{a}\frac{1}{64x^2}\ \mathrm{d}x$ Ignore limits
A1	For $=\pi\left(\frac{1}{16}-\frac{1}{64a}\right)$
M1	Dependant on at least one previous M mark being awarded - for subtraction of the two integrals.
A1	For $32a^3-33a+1=0$ oe
A1	For $a=1$
	Alternative
M1	For $\pi\int_{\frac{1}{4}}^{a}\left(x-\frac{1}{64x^2}\right)\mathrm{d}x=\frac{27\pi}{64}$ Ignore limits
A1	For $\int_{\frac{1}{4}}^{a}\left(64x-\frac{1}{x^2}\right)\mathrm{d}x=27$ Ignore limits
M1	For an attempt to integrate $\int_{\frac{1}{4}}^{a}\left(64x-\frac{1}{x^2}\right)\mathrm{d}x$ Ignore limits
A1	$\left[32x^2+\frac{1}{x}\right]_{\frac{1}{4}}^{a}=27$
M1	Dependant on at least one previous M mark being awarded - for correct substitution of the limits
A1	For $32a^3-33a+1=0$ oe
A1	For $a=1$

Type MS · Factorising and solving a cubic

Answer 15 of 22 · Question 4 part c · 2 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 4 · find part c in this mark scheme extract

Number		
4 (a)	$f(2)=2\times2^3+a\times2^2+b\times2+18=0$ $f'(x)=6x^2+2ax+b \Rightarrow f'(2)=6\times2^2+2\times a\times2+b=5$ $4a+2b+34=0$ $4a+b+19=0$ $\Rightarrow b=-15,\ a=-1$	M1 M1M1 A1 M1A1 [6]
(b)	$x-2\overline{)2x^3-x^2-15x+18}$ $2x^2+3x-9$ $2x^2+3x-9=(x+3)(2x-3)$ $\Rightarrow (x-2)(x+3)(2x-3)$	M1 M1A1 [3]
(c)	$x=2,\ -3,\ \frac{3}{2}$	B2ft [2]
		Total 11 marks

Type MS · Factorising and solving a cubic

Answer 15 of 22 · Question 4 part c · 2 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 4 · find part c in this mark scheme extract

Question number	Scheme	Marks
3(a)	$\frac{dy}{dx} = 2e^{2x}\left(x^2+1\right)+e^{2x}\left(2x\right)$	M1A1A1 [3]
(b)	When $x=0$ $\frac{dy}{dx} = 2\times1\times1+1\times0=2$ $\quad y=e^{2\times0}\left(0+1\right)=1$ $y-1=2\left(x-0\right)\Rightarrow y=2x+1$	B1B1 B1 [3]
		Total 6 marks

(a)	
M1	Attempted use of the product rule. Sum of two terms (either way round) with $x^n \to x^{n-1}$ (Condone e^{2x} instead of $2e^{2x}$) Once the correct answer is seen ISW. This mark may be implied by the sum of two terms with one of the two terms correct.
A1	Either term correct
A1	Both terms correct
(b)	
B1	When $x=0$ $\frac{dy}{dx}=2$
B1	When $x=0$ $y=1$
B1	$y=2x+1$

Type MS · Factorising and solving a cubic

Answer 15 of 22 · Question 4 part c · 2 marks

2020 JAN · Paper 01 (regional R) · extract 3 of 4 · find part c in this mark scheme extract

Question number	Scheme	Marks
4 (a)	$f(2)=2\times2^3+a\times2^2+b\times2+18=0$ $f'(x)=6x^2+2ax+b \Rightarrow f'(2)=6\times2^2+2\times a\times2+b=5$ $4a+2b+34=0$ $4a+b+19=0$ $\Rightarrow b=-15,\ a=-1$	M1 M1M1 A1 M1A1 [6]
(b)	$\begin{array}{r} 2x^2+3x-9 \\ x-2\overline{)2x^3-x^2-15x+18} \end{array}$ $2x^2+3x-9=(x+3)(2x-3)$ $\Rightarrow (x-2)(x+3)(2x-3)$	M1 M1A1 [3]
(c)	$x=2,\ -3,\ \frac{3}{2}$	B2ft [2]
		Total 11 marks

(a)	
M1	$f(2)=0$ leading to an equation in a and b
M1	Attempt to differentiate
M1	$f'(2)=5$ leading to an equation in a and b
A1	$4a+2b+34=0$ and $4a+b+19=0$
M1	Solving simultaneously
A1	$b=-15,\ a=-1$
(b)	
M1	Dividing by $x-2$ to obtain a 3TQ
M1	Factorising the 3TQ
A1	All 3 terms correct
(c)	
B2 ft	$x=2,\ -3,\ \frac{3}{2}$ (**B1** for 2 correct)

Type MS · Factorising and solving a cubic

Answer 15 of 22 · Question 4 part c · 2 marks

2020 JAN · Paper 01 (regional R) · extract 4 of 4 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 16 of 22 · Question 4 part c · 2 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

4(a)	$f'(x) = 3px^2 + 2qx - 37$ $f'(-2) = 3p(-2)^2 + 2q(-2) - 37 = -33$ $f(-5) = p(-5)^3 + q(-5)^2 - 37(-5) - 12q = 0$ $\begin{cases} 12p - 4q = 4 \\ -125p + 13q = -185 \end{cases} \Rightarrow p = \cdots \; q = \cdots$ $p = 2, q = 5$	M1 M1 M1 dM1 A1A1 [6]
(b)	$x+5\overline{)2x^3+"5"x^2-37x-"60"}$ $\quad 2x^2-"5"x-"12"$ OR $2x^3 + "5"x^2 - 37x - "60" = (x+5)(Ax^2 + Bx + C)$ $\Rightarrow A = 2, B = \cdots, C = \cdots$ $2x^2 - 5x - 12 = (2x+3)(x-4)$ $\{f(x) =\}(x+5)(2x+3)(x-4)$	M1 M1 A1 [3]
(c)	$(x+5)(2x+3)(x-4) = 0, \Rightarrow x = -5,\ 4,\ -\frac{3}{2}$	M1A1 [2]
		Total 11 marks

Part	Mark	Notes
(a) Mark i and ii together	**M1**	For differentiating the given expression in terms of p and q with respect to x, at least one term correct and no terms integrated.
	M1	Attempts $f'(\pm 2) = -33$
	M1	Attempts $f(\pm 5) = 0$
	dM1	For solving the two resulting simultaneous equations by any correct method (allow slips) providing there are two variables p and q in both, leading a value of p or q. Explicit method to solve must been seen. **Depends on all previous method marks**
	A1	For $p = 2$ or $q = 5$ from correct working in two correct equations
	A1	For both $p = 2$ and $q = 5$ from correct working
(b)	**M1**	For using polynomial division (or any other **algebraic** method) to find the quadratic factor. **Division:** Look for divisor of $(x+5)$, quotient of a quadratic $2x^2 -$ their $qx + c$ $(c \neq 0)$ **Compare coefficients**: minimum required is an identity of the correct form followed by values of $A = 2, B = -$their $q, C = \ldots (C \neq 0)$
	M1	Correct method of factorising their quadratic factor. (See General Principles)
	A1	Correct factorised expression written in full on one line.
(c)	**M1**	Correct method of solving their $f(x) = 0$. Their $f(x)$ must have 3 solutions, one of which must be $x = -5$, "=0" or their method may be implied by their x values. They must use their result in part b which $f(x)$ reaches 2 or 3 factors, if part b is not attempted, a full algebraic method of solving $f(x)=0$ must be seen in this part.
	A1	For all three correct values of x, no extra solutions.

Type MS · Factorising and solving a cubic

Answer 16 of 22 · Question 4 part c · 2 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

(a) **ALT I Algebraic Division**

M1 Same as main scheme
M1 Their f′(*x*) is in the form of a 3TQ, look for divisor of (*x*+2), quotient of a linear expression leading to a constant remainder in $\boldsymbol{p}$ and $\boldsymbol{q}$, sets = -33
M1 For dividing f(*x*), look for divisor of (*x*+5), quotient of a 3TQ, leading to a constant remainder in $\boldsymbol{p}$ and $\boldsymbol{q}$, sets=0
dM1A1 Same as main scheme

You may also see variations for their division, marking is similar as above, (if not sure, send to review)
e.g.
For f′(*x*)

$$\begin{array}{r|lll} -2 & 3p & 2q & -37 \\ & & -6p & 12p-4q \\ \hline & 3p & -7p+2q & 12p-4q-37=-33 \end{array}$$

For f(*x*)

$$\begin{array}{r|llll} -5 & p & q & -37 & -12q \\ & & -5p & -5q+25p & 25q-125p+185 \\ \hline & p & q-5p & -5+25p-37 & 13q-125p+185=0 \end{array}$$

(a) **ALT II Compare coefficients**
M1 Same as main scheme
M1 Sets their f′(*x*) which is in the form of a 3TQ, equal to (*x*+2)(K*x*+C)-33 ($K\neq 0, C\neq 0$) , then attempts to equate the coefficients of *x* to form an equation in $\boldsymbol{p}$ and $\boldsymbol{q}$
M1 Sets f(*x*)=(*x*+5) $[\boldsymbol{p}x^2+Rx+S]$ $(R\neq 0, S\neq 0)$, then attempts to equate their coefficients of *x* to form an equation in $\boldsymbol{p}$ and $\boldsymbol{q}$.
dM1A1 Same as main scheme

For reference, the correct working for this method:

$$\begin{aligned} \mathrm{f}'(x) &= 3px^2+2qx-37 \\ &= (x+2)(3px-2)-33 \\ \Rightarrow 6px-2x &= 2qx \\ 6p-2 &= 2q \end{aligned}$$

$$\mathrm{f}(x)=px^3+qx^2-37x-12q=(x+5)\left[px^2+(q-5p)x-\frac{12q}{5}\right]$$

$$\begin{aligned} \Rightarrow 5(q-5p)x-\frac{12q}{5}x &= -37x \\ 5(q-5p)-\frac{12q}{5} &= -37 \\ \frac{13q}{5}-25p &= -37 \end{aligned}$$

Type MS · Factorising and solving a cubic

Answer 16 of 22 · Question 4 part c · 2 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 17 of 22 · Question 7 part b · 3 marks

2025 NOV · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

7(a)	$\left[\mathrm{f}\left(-\frac{1}{2}\right)=\right]2\times\left(-\frac{1}{2}\right)^3+A\times\left(-\frac{1}{2}\right)^2+B\times\left(-\frac{1}{2}\right)-20=0\left[\Rightarrow A-2B=81\right]$	M1
	$\left[\mathrm{f}'(x)=\right]6x^2+2Ax+B$	M1
	$\left[\mathrm{f}'(1)=\right]6\times1^2+2A\times1+B=-27\left[\Rightarrow 2A+B=-33\right]$	M1
	$A-2B=81$ oe	
	$2A+B=-33$ oe	dddM1
	$\Rightarrow A=3^*,\quad B=-39$	A1 cso [5]
(b)	$2x+1\overline{)2x^3+3x^2-39x+20}$ $\quad x^2+x-20$	M1
	$x^2+x-20=(x-4)(x+5)$	dM1
	$\Rightarrow\left[x=\right]-\frac{1}{2},4,-5$	A1 [3]
(c)	$\left[A=\right]\int_{"-5"}^{"-\frac{1}{2}"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x+\left\lvert\int_{"-\frac{1}{2}"}^{"4"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x\right\rvert$	M1
	oe eg $\int_{"-5"}^{"-\frac{1}{2}"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x+\int_{"4"}^{"-\frac{1}{2}"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x$	
	$\left[A=\right]\left[\frac{2x^4}{4}+\frac{3x^3}{3}-\frac{39x^2}{2}-20x\right]_{[-5]}^{\left[-\frac{1}{2}\right]}+\left[\frac{2x^4}{4}+\frac{3x^3}{3}-\frac{39x^2}{2}-20x\right]_{\left[-\frac{1}{2}\right]}^{[4]}$	M1
	$\left[A=\right]\left\{\left[\frac{2\left(-\frac{1}{2}\right)^4}{4}+\frac{3\left(-\frac{1}{2}\right)^3}{3}-\frac{39\left(-\frac{1}{2}\right)^2}{2}-20\left(-\frac{1}{2}\right)\right]-\left[\frac{2(-5)^4}{4}+\frac{3(-5)^3}{3}-\frac{39(-5)^2}{2}-20(-5)\right]\right\}$	
	$\left[+\right]\left\{\left[\frac{2(4)^4}{4}+\frac{3(4)^3}{3}-\frac{39(4)^2}{2}-20(4)\right]-\left[\frac{2\left(-\frac{1}{2}\right)^4}{4}+\frac{3\left(-\frac{1}{2}\right)^3}{3}-\frac{39\left(-\frac{1}{2}\right)^2}{2}-20\left(-\frac{1}{2}\right)\right]\right\}$ oe	M1
	$\left[A=\right]\frac{6561}{16}$ or 410.0625 oe	A1 [4]
		Total 12 marks

Type MS · Factorising and solving a cubic

Answer 17 of 22 · Question 7 part b · 3 marks

2025 NOV · Paper 02 · extract 2 of 4 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	Substitutes $-\frac{1}{2}$ into f(x) and sets = 0 Accept as a minimum $2\times\left(-\frac{1}{8}\right)+A\times\frac{1}{4}+B\times\left(-\frac{1}{2}\right)-20=0$ or $-\frac{1}{4}+A\times\frac{1}{4}+B\times\left(-\frac{1}{2}\right)-20=0$
	M1	For correctly differentiating f(x)
	M1	Substitutes 1 into their derivative, and sets = – 27 Accept $2A+B=-33$ with no errors. This isn't a dependent mark, but candidates must be substituting into a changed expression with at least 2 terms in x
	dddM1	Solves their linear simultaneous equations by any method, allow one error in solving. Minimal working here is fine, but there must be **some** working. Candidates who show no working to solve the simultaneous equations will not gain this mark (and therefore not the next mark) As this is a 'show' question, correct answers cannot imply this method mark. Dependent on previous 3 method marks.
	A1*cso	For the correct value of A [which is given in the question] and the correct value of B. Minimum steps as shown/stated, no errors or omissions.
(b)	M1	Award the mark if the candidate is seen to be attempting any method, such as algebraic division or equating coefficients (note this will often have minimal working) and attains: $x^2\pm Cx-20$ or $x^2+x\pm D$ $C,D\neq 0$ It is not sufficient to show simply a product of linear factors.
	dM1	For correctly factorising their quadratic factor or correctly using any method to solve their quadratic factor = 0 Dependent on previous method mark.
	A1	For all three roots. Candidates are told to use algebra, so correct answers alone cannot imply M marks for this question. This means correct answers alone will be 0 marks overall.
The majority of candidates are completing this question, using the factor of $(2x + 1)$ given. Occasionally, we are seeing a re-start using a different factor – probably deduced from a calculator, but if sufficient algebra is shown, we will award the marks. For $(x + 5)$ need to see $2x^2-7x-4$ with $2x^2$ correct and either the – 7 or – 4 correct For $(x – 4)$ need to see $2x^2+11x+5$ with $2x^2$ correct and either the +11 or +5 correct for the first M mark – the next 2 marks then follow.		

Type MS · Factorising and solving a cubic

Answer 17 of 22 · Question 7 part b · 3 marks

2025 NOV · Paper 02 · extract 3 of 4 · find part b in this mark scheme extract

(c)	M1	For a correct statement of the required area. Allow use of their limits from part (b), even if working wasn't shown. However, no limit may be 0 This mark may be implicit in further working and may even be awarded at the stage when a negative numerical area has been made positive.
	M1	For a minimally acceptable attempt at integration of an expression of the form $2x^3+3x^2+Px-20$ $P\neq 0$. The modulus sign and the addition between the 2 calculations do not need to be seen for this mark. See general guidance, additionally, no power of x to decrease. Limits do no need to be present for this mark.
	M1	For a full and correct substitution of each of their limits (their roots from part (b), no limit 0) into any changed expression with a minimum of 2 terms. The modulus sign and the addition between the 2 calculations do not need to be seen for this mark. A correct final answer, if the integration is shown, may imply this mark.
	A1	For the correct exact area. The correct final answer may imply the first and third method marks, but may not imply the second method mark (ie must show integration). If no integration is shown, with a correct answer, the only possible marks are M1 M0 M1 A0 (if it's clear somewhere in working that the candidate has split the area). or M0 M0 M1 A0 (if the answer is correct with no reference to splitting the area).
If candidates work with $\int_{"-5"}^{"4"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x$ they **may** attain up to M0 M1 M1 A0		

Type MS · Factorising and solving a cubic

Answer 17 of 22 · Question 7 part b · 3 marks

2025 NOV · Paper 02 · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 18 of 22 · Question 11 part b · 6 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

11 (a)	$f(x)=\int ax^2-14x-10\,dx=\frac{ax^3}{3}-\frac{14x^2}{2}-10x+c$	M1A1
	$f(4)=\frac{a\times4^3}{3}-\frac{14\times4^2}{2}-10\times4+c=0\Rightarrow\frac{64a}{3}-152+c=0$	M1
	$f(-1)=\frac{a\times(-1)^3}{3}-\frac{14\times(-1)^2}{2}-10\times(-1)+c=25\Rightarrow-\frac{a}{3}-22+c=0$	M1
	$\frac{64a}{3}-152+c=0$ $-\frac{a}{3}-22+c=0$	
	$\Rightarrow 152-\frac{64a}{3}=22+\frac{a}{3}\Rightarrow a=6$	M1A1 [6]
(b)	$c=22+\frac{6}{3}=24$	B1
	$f(x)=2x^3-7x^2-10x+24$	
	$x-4\overline{)2x^3-7x^2-10x+24}$ $\quad 2x^2+x-6$	M1A1
	$f(x)=(x-4)(2x-3)(x+2)=0$	dM1A1
	$x=4,\ \frac{3}{2},\ -2$	A1 [6]
	ALT $2x^3-7x^2-10x+24=(x-4)(ax^2+bx+c)$ $a=2, b=1, c=-6$	[M1
	$f(x)=(x-4)(2x^2+x-6)$	A1
	$f(x)=(x-4)(2x-3)(x+2)=0$	dM1A1
	$x=4,\frac{3}{2},-2$	A1]
		Total 12 marks

Type MS · Factorising and solving a cubic

Answer 18 of 22 · Question 11 part b · 6 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

	M1	For correct method to solve the equations simultaneously to find a A correct intermediate step is required e.g. $65a = 390$
	A1 cso	For fully correct working leading to $a = 6$
(b)	**B1**	For correctly identifying $c = 24$
	M1	For attempt to divide f(x) by $x - 4$ Must get as far as $2x^2 + \cdots$
	A1	For correct division of f(x) by $x - 4$ to obtain $2x^2 + x - 6$
	dM1	For an attempt to factorise their 3TQ which must come from a cubic. See General Guidance on what constitutes an attempt to factorise.
	A1	For obtaining correct factorisation of the cubic: $(x - 4)(2x - 3)(x + 2)$
	A1	For all three correct solutions of the equation: $x = 4, \frac{3}{2}, -2$
Alternative method		
	M1	For an attempt to find the quadratic factor that multiplies $x - 4$ to give f(x) Must get as far as $f(x) = (x - 4)(2x^2 + bx + c)$
	A1	For correctly comparing coefficients to obtain $2x^2 + x - 6$
	dM1	For an attempt to factorise their 3TQ which must come from a cubic. See General Guidance on what constitutes an attempt to factorise.
	A1	For obtaining correct factorisation of the cubic: $(x - 4)(2x - 3)(x + 2)$
	A1	For all three correct solutions of the equation: $x = 4, \frac{3}{2}, -2$
	Note: Correct solution seen with no working scores B0M0A0M0A0A0	

Type MS · Factorising and solving a cubic

Answer 18 of 22 · Question 11 part b · 6 marks

2021 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

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Type MS · Factorising and solving a cubic

Answer 19 of 22 · Question 7 part b · 4 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

7 (a)	$64\left(\frac{1}{4}\right)^3 - 64\left(\frac{1}{4}\right)^2 + 3 = 0$ so $(4x-1)$ is a factor	M1 A1 [2]
(b)	$16x^2 - 12x + "-3"$ $4x-1 \overline{)64x^3 - 64x^2 \{+0x\} + 3}$ $64x^3 - 16x^2$ $-48x^2$ or $64x^3 - 64x^2 + 3 = (4x-1)(16x^2 - 12x + C)$ $x = \frac{--12 \pm \sqrt{(-12)^2 - 4(16)(-3)}}{2\times 16}$ $x = \frac{1}{4}$ oe $x = \frac{3 \pm \sqrt{21}}{8}$ oe	M1 dM1 B1 A1 [4]
(c)	$ar^2 = 9$ or $\frac{a}{1-r} = 192$ oe $(192 - 192r)r^2 = 9$ oe $192r^2 - 192r^3 = 9 \Rightarrow 64r^3 - 64r^2 + 3 = 0$	B1 M1 A1*cso [3]
(d)	$r = \frac{1}{4}$	B1 [1]
(e)	$a\left(\frac{1}{4}\right)^2 = 9$ or $\frac{a}{1-\frac{1}{4}} = 192$ $\Rightarrow a = ...$ $a = 144$	M1 A1*cso [2]
(f)	$\frac{144\left(1-0.25^n\right)}{1-0.25} > 191.9$ $1 - 0.25^n > \frac{1919}{1920}$ $0.25^n < 0.00052083...$ $n > \frac{\log 0.00052083}{\log 0.25}$ $n > 5.45... \Rightarrow n = 6$	M1 dM1 M1 A1 [4]
		Total 16 marks

Type MS · Factorising and solving a cubic

Answer 19 of 22 · Question 7 part b · 4 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For $64\left(\pm\frac{1}{4}\right)^3-64\left(\pm\frac{1}{4}\right)^2+3$
	A1	For = 0 and a conclusion stated, no errors seen. Accept minimal conclusion.
(b)	M1	Attempts long division. Minimally acceptable attempt is the division and correct working as written in the scheme. Must get a 3TQ factor. If comparing coefficients, a correct equation/comparison must be written followed by an attempt to find C. Must get a 3TQ factor.
	dM1	For a fully correct method to solve their 3TQ to find x, correct solutions from a correct 3TQ implies this mark.
	B1	For correct root $\frac{1}{4}$
	A1	For correct roots $\frac{3\pm\sqrt{21}}{8}$
(c)	B1	For either correct equation.
	M1	For both correct equations and any correct step to eliminate a
	A1*cso	For a fully correct solution, no errors, with at least one correct intermediate line leading to $64r^3-64r^2+3=0$
(d)	B1 **(A1 on ePen)**	For the correct value of r
(e)	M1	For correct substitution of their value for r in either of their equations to find a
	A1	For $a = 144$
(f)	M1	**Allow working in equation or any inequality for all M marks** Correct use of sum to n terms formula with $a = 144$ or their a and their r to set up an equation or inequality
	dM1	Attempts to rearrange and obtains $\pm(\text{"their } r\text{"})^n \ldots p$
	M1	Correct use of logs and attempt to evaluate the log. (Can be implied by awrt 5.5 only)
	A1	$n = 6$ (must come from solving inequalities with no errors in processing the inequalities. If from solving equations only, award A0)

Type MS · Factorising and solving a cubic

Answer 19 of 22 · Question 7 part b · 4 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 20 of 22 · Question 3 part b · 3 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

3(a)	$\left(y=\int\left(mx^2-10x-37\right)\mathrm{d}x=\right)\frac{mx^3}{3}-\frac{10x^2}{2}-37x(+c)$	M1
	$\left[\text{At } (1, 20)\right]$ $20=\frac{m\times1^3}{3}-\frac{10\times1^2}{2}-37\times1+c\left(\Rightarrow 62=\frac{m}{3}+c\right)$ oe	dM1
	$\left[\text{Factor of } (x-5)\right]$ $0=\frac{m\times5^3}{3}-\frac{10\times5^2}{2}-37\times5+c\left(\Rightarrow 310=\frac{125m}{3}+c\right)\Rightarrow m=6 \quad c=60$	M1 M1
	$\Rightarrow\left(\mathrm{g}(x)=\right)2x^3-5x^2-37x+60$ *	A1cso [5]
ALT	3rd and 4th method marks $(x-5)\left(\frac{m}{3}x^2+bx\mp\frac{c}{5}\right)=\frac{m}{3}x^3-5x^2-37x\pm c\Rightarrow m=6 \quad c=60 \quad (b=5)$	M1 M1
Special case up to SC3 – mark as M1dM1M1 M0A0 in Epen	Differentiates the given expression for g (x)correctly to give $2(3x^2)-5(2x)-37\left(=6x^2-10x-37\right)$ **and** concludes $m = 6$	SC1
	Correctly substitutes $x = 5$ into the given function and shows g (x) = 0 $\left(\mathrm{g}(5)=\right)2\times5^3-5\times5^2-37\times5+60=0$	SC1
	Correctly substitutes $x = 1$ into the given function and shows and g (x) = 20 $\left(\mathrm{g}(1)=\right)2\times1^3-5\times1^2-37\times1+60=20$	SC1
(b)	$x-5\overline{)2x^3-5x^2-37x+60}$ $\quad 2x^2+5x-12$	M1
	$2x^2+5x-12=(2x-3)(x+4)$ $\left(\mathrm{g}(x)=(x-5)(2x-3)(x+4)=[0]\right)$	M1
	$\Rightarrow x=5,\ \frac{3}{2},\ -4$	A1 [3]
		Total 8 marks

Type MS · Factorising and solving a cubic

Answer 20 of 22 · Question 3 part b · 3 marks

2023 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For a minimally acceptable attempt to integrate the given expression (see general guidance). No power of x to decrease. $+c$ is not required for this mark
	dM1	Correctly substitutes $x = 1$ and $y = 20$ to form an equation – simplification is not required Dependent on 1st method mark. A fully correct equation with no incorrect working can imply this mark. We will be lenient on not seeing the substitution of $x = 1$ as this is trivial, so long as the resulting equation is correct from their g (1) = 20. $+c$ is not required for this mark
	M1	Uses the information correctly that $(x – 5)$ is a factor of g (x) by correct substitution of $x = 5$ and $g(x) = 0$ into their g (x). Their g (x) **must** be a 4 term cubic expression. A fully correct equation with no incorrect working can imply this mark – but if the equation is incorrect we must see substitution of $x = 5$ such that $g(5) = 0$.
	M1	Uses a valid and complete method to solve their resulting simultaneous equations which must be in two variables. Allow up to 2 errors.
	A1 cso	For the correct function as shown or the correct expression. This is a show question. There must be no errors for the award of this mark.
ALT	3rd M1	Uses the information correctly that $(x – 5)$ is a factor of g (x) by writing a correct statement like the one shown for their g (x). Their g (x) **must** be a 4 term cubic expression.
	4th M1	Uses a valid and complete method to solve their resulting simultaneous equations in three variables. Allow up to 2 errors. It is not necessary to see the value for "b".
(b)	M1	For an attempt at polynomial division or equate coefficients to reach a 3 term quotient. The minimally acceptable quotient is $2x^2 \pm 5x \pm k \quad k \neq 0$ where k is a constant. In the absence of any other working, $2x^2 \pm 5x \pm k \quad k \neq 0$ may be accepted as evidence of working.
	M1	For attempting to factorise their quotient which must be of the form $2x^2 \pm 5x \pm k \quad k \neq 0$ - we must at least see a quadratic factor to factorise. See General Guidance for the definition of an attempt. It would also be possible to see a minimal attempt to use the quadratic formula or completing the square to solve (see general guidance) if their quadratic, of the form $2x^2 + 5x \pm k \quad k \neq 0$ does not factorise. If using the formula the substitution of their a, b and c must be completely correct.
	A1	For the correct values of x
Candidates have clearly been directed to use algebra in the question and therefore 0 marks can be scored without seeing M1 M1		

Type MS · Factorising and solving a cubic

Answer 20 of 22 · Question 3 part b · 3 marks

2023 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 21 of 22 · Question 8 part b · 6 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

8(a)	$\int 17+2x-3x^2\ \mathrm{d}x = 17x+\frac{2x^2}{2}-\frac{3x^3}{3}+k$	M1A1
	$0=\left(17\times(-1)+\frac{2\times(-1)^2}{2}-\frac{3\times(-1)^3}{3}\right)+k \Rightarrow k=15$	M1
	$y=15+17x+x^2-x^3$	A1 [4]
(b)	$\frac{(15+17x+x^2-x^3)}{(x+1)}=-x^2+2x+15$	M1A1
	$-x^2+2x+15=(x+3)(5-x)$	M1A1
	$a=-3,\ [-1]$ and $b=5$	A1
	When $x=0$ $y=15$ so $c=15$	B1 [6]
(c)	$\int_0^5(15+17x+x^2-x^3)\ \mathrm{d}x-\frac{1}{2}\times5\times15$	M1
	OR	
	$\int_0^5(15+17x+x^2-x^3)\ \mathrm{d}x-\int_0^5(15-3x)\ \mathrm{d}x$	
	$\int_0^5(15+17x+x^2-x^3)\ \mathrm{d}x=\left[15x+\frac{17x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\right]_0^5$	M1
	$\left(15\times5+\frac{17\times5^2}{2}+\frac{5^3}{3}-\frac{5^4}{4}\right)-(0)=\left(\frac{2075}{12}\right)$	M1
	Area of triangle Area $=\frac{1}{2}\times5\times15=37.5$ **OR**	B1
	Area $=\int_0^5(15-3x)\ \mathrm{d}x=\left[15x-\frac{3x^2}{2}\right]_0^5=37.5$	[B1]
	For the correct area of $R=\frac{2075}{12}-37\frac{1}{2}=\frac{1625}{12}=135\frac{5}{12}$	A1 [5]
		Total 15 marks

Type MS · Factorising and solving a cubic

Answer 21 of 22 · Question 8 part b · 6 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

Question	Notes	Marks
8(a)	$f'(x)=17+2x-3x^2$	
	For an attempt to integrate $f'(x)$ $y=17x+\frac{2x^2}{2}-\frac{3x^3}{3}+(k)$	M1
	For the correct integral **including** a constant of integration $y=17x+\frac{2x^2}{2}-\frac{3x^3}{3}+k$	A1
	For substituting $(-1\ ,\ 0)$ into their integrated expression, which must include a constant of integration. $0=\left(17\times(-1)+\frac{2\times(-1)^2}{2}-\frac{3\times(-1)^3}{3}\right)+k\Rightarrow(k=15)$	M1
	For writing the equation in the required form $y=15+17x+x^2-x^3*$ **This is a given equation. Every step above must be seen for the award of full marks.**	A1 cso [4]
(b)	Divides $(15+17x+x^2-x^3)$ by $(x+1)$ $x+1\overline{)15+17x+x^2-x^3}\quad Q+2x+x^2$ **OR** Equates coefficients $(15+17x+x^2-x^3)=(x+1)(Ax^2+Bx+c)\Rightarrow(x+1)(-x^2+2x+Q)$ Where Q is a constant **Minimal working here is sight of the quadratic factor.**	M1
	For obtaining the correct 3TQ $-x^2+2x+15$	A1
	For factorising their 3TQ [or otherwise solving] $-x^2+2x+15=(x+3)(5-x)$	M1A1
	For $a=-3$, $[-1]$ and $b=5$ identified clearly.	A1
	SC No working seen [use of a root finder on a calculator]	
	For $(x+1)(x-5)(x+3)$ seen leading to $a=-3$ and $b=5$ with no working award: M0A0MA1A1	
	For $a=-3$ and $b=5$ seen with no other working seen award: M0A0M0A0A1	
	For the value of $c=15$	B1
(c)	For writing a correct expression for the area of R with the correct limits $\int_0^5(15+17x+x^2-x^3)\,dx-\frac{1}{2}\times5\times15$ OR	M1

Type MS · Factorising and solving a cubic

Answer 21 of 22 · Question 8 part b · 6 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

	$\int_0^5 \left(15+17x+x^2-x^3\right)\,\mathrm{d}x-\int_0^5 (15-3x)\,\mathrm{d}x$	
	For an attempt to integrate the expression for the curve. [Ignore limits for this mark] Area $= \int_0^5 \left(15+17x+x^2-x^3\right)\,\mathrm{d}x = \left[15x+\frac{17x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\right]_0^5$	M1
	For evaluating their integral using their limits. Area $= \left(15\times5+\frac{17\times5^2}{2}+\frac{5^3}{3}-\frac{5^4}{4}\right)-(0)=\left(\frac{2075}{12}\right)$	M1
	For the area of the triangle Area $=\frac{1}{2}\times5\times15=37.5$ **ALT** Integrates the line Area $= \int_0^5 (15-3x)\,\mathrm{d}x=\left[15x-\frac{3x^2}{2}\right]_0^5=37.5$	B1 [B1]
	For the correct area of R $\frac{2075}{12}-37\frac{1}{2}=\frac{1625}{12}=135\frac{5}{12}$	A1 [5]
	ALT	
	For writing an expression for the area of R with the correct limits. $\int_0^5 \left(15+17x+x^2-x^3\right)\,\mathrm{d}x-\int_0^5 (15-3x)\,\mathrm{d}x=\left[\int_0^5 \left(20x+x^2-x^3\right)\,\mathrm{d}x\right]$ Award the B mark for a correct expression for the combined area.	M1 B1
	For an attempt to integrate the expression for the combined area or just the curve. [Ignore limits for this mark] Area $= \int_0^5 \left(20x+x^2-x^3\right)\,\mathrm{d}x=\left[\frac{20x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\right]_0^5$	M1
	For evaluating their integral using their limits, but the lower limit must be 0. Area $=\left(\frac{20\times5^2}{2}+\frac{5^3}{3}-\frac{5^4}{4}\right)=\left(\frac{1625}{12}\right)$	M1
	For the correct final area.	A1 [5]
		Total 15 marks

Type MS · Factorising and solving a cubic

Answer 21 of 22 · Question 8 part b · 6 marks

2023 JAN · Paper 01 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Factorising and solving a cubic

Answer 22 of 22 · Question 9 part c · 5 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

9 a	$(ax^4 = 3x^4) \Rightarrow a = 3$	B1
	$(4c = 64) \Rightarrow c = 16$	B1
	$(bx^3 - 4ax^3 = 4x^3) \Rightarrow b - 4a = 4$ or $(4ax^2 - 4bx^2 + cx^2 = -36x^2)$ $\Rightarrow 4a - 4b + c = -36$ *or* $(4bx - 64x = 0x) \Rightarrow 4b - 64 = 0$	M1
	$b = 16$	A1 (4)
b	$\int_0^2 (x^3 + x^2 - 6x)\mathrm{d}x = \left[\frac{x^4}{4} + \frac{x^3}{3} - \frac{6}{2}x^2\right]_0^2$	B1 (M1 on ePen) M1
	$= \left("\frac{2^4}{4} + \frac{2^3}{3} - 3(2)^2"\right) - (0)$	M1 (A1 on ePen)
	$= \pm\frac{16}{3}$ oe	A1 (M1 on ePen)
	$\left(\left["\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right]_x^0 = \pm\frac{16}{3} \Rightarrow\right) (0) - \left("\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right) = \pm\frac{16}{3}$ oe	M1
	$3x^4 + 4x^3 - 36x^2 + 64 = 0$	A1
	$(x-2)^2(3x^2 + 16x + 16) = 0$ *	A1* cso (7)
ALT	$\int_x^2 (x^3 + x^2 - 6x)\mathrm{d}x = \left["\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right]_x^2$	B1 (M1 on ePen) M1
	$= \left("\frac{2^4}{4} + \frac{2^3}{3} - 3(2)^2"\right) - \left("\frac{x^4}{4} + \frac{x^3}{3} - 3(x)^2"\right)$	M1 (A1 on ePen)
	$\pm\left(-\frac{16}{3} - \frac{x^4}{4} - \frac{x^3}{3} + 3x^2\right) = 0$	A1 (M1 on ePen)
	$3x^4 + 4x^3 - 36x^2 + 64 = 0$	M1 A1
	$(x-2)^2(3x^2 + 16x + 16) = 0$ *	A1
c	(When $(x-2) = 0 \quad x = 2$ and when $3x^2 + 16x + 16 = 0$) $(3x+4)(x+4) = 0 \qquad$ so $\qquad x = -\frac{4}{3}$ and $x = -4$	M1 A1
	($x \neq -4$ as it is to the left of the point at $x = -3$) $x = -\frac{4}{3}$	M1
	When "$x = -\frac{4}{3}$" $\quad y = \left("-\frac{4}{3}"\right)^3 + \left("-\frac{4}{3}"\right)^2 - 6\left("-\frac{4}{3}"\right) = \frac{200}{27}$	dM1
	So $A = \left(-\frac{4}{3}, \frac{200}{27}\right)$	A1 (5)
		Total 16 marks

Type MS · Factorising and solving a cubic

Answer 22 of 22 · Question 9 part c · 5 marks

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Part	Mark	Notes
(a)	B1	$a = 3$
	B1	$c = 16$
	M1	For a clear process to equate coefficients to find either of the equations $b - 4a = 4$ or $4a - 4b + c = -36$ or $4b - 64 = 0$. Allow one error. For algebraic division: there must be a complete attempt to divide by $x^2 + px + q$ and reach an expression of the form $3x^2 + rx + 16,\ p, q, r \neq 0$
	A1	$b = 16$
Full marks may be awarded for all of *a, b* and *c* appearing correctly with no method shown. Embedded *a, b* and *c* should be awarded full marks.		
(b)	B1 (M1 on ePen)	Correct limits of $x = 0$ and $x = 2$ – or used later in working.
	M1	For a minimally acceptable attempt to integrate any 3 term cubic of the form $x^3 + fx^2 + gx$. Limits do not need to be present. See general guidance for the definition of a minimally acceptable attempt.
	M1 (A1 on ePen)	Substitution of their limits into any changed expression, the correct way round minimum 3 terms. Sub of 0 not needed. Allow this mark to be implied by a fully correct value.
	A1 (M1 on ePen)	Either for a fully correct substitution, or for $\pm\frac{16}{3}$
	M1	For $(0) - \left("\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right) = \pm\frac{16}{3}$ oe. Allow use of their integrated expression and their $\pm\frac{16}{3}$ which can be in any equivalent form, including an unevaluated expression. Allow use of *n* rather than *x*. The limit of 0 need not be seen.
	A1	For $3x^4 + 4x^3 - 36x^2 + 64 = 0$ Allow use of *n* rather than *x*. $\frac{16}{3}$ must have been evaluated.
	A1* cso	For $(x-2)^2(3x^2 + 16x + 16) = 0$ .No errors in working. Candidates can use *n* throughout.
ALT	B1 (M1 on ePen)	Correct limits of $x = x$ and $x = 2$ – seen on the integral or used later in working. These can be either way round.
	M1	For a minimally acceptable attempt to integrate any 3 term cubic of the form $x^3 + fx^2 + gx$. Limits do not need to be present. See general guidance for the definition of a minimally acceptable attempt.
	M1 (A1 on ePen)	For substitution of both of their limits into any changed expression. Minimum 3 terms
	A1 (M1 on ePen)	For $\pm\left(-\frac{16}{3} - \frac{x^4}{4} - \frac{x^3}{3} + 3x^2\right) = 0$
	M1	For a valid attempt to multiply their equation throughout to arrive at integer coefficients.
	A1 A1	As main scheme. Candidates can use *n* throughout.
(c)	M1	For a complete and minimally acceptable attempt to solve $3x^2 + 16x + 16 = 0$ to give $x =$ See general guidance for definition of minimally acceptable. Some candidates may have factorised in part b), if used in part c), marks can be awarded
	A1	For $x = -\frac{4}{3}$ and $x = -4$
	M1	Chooses $"x = -\frac{4}{3}"$, can be stated explicitly or used later implicitly. Allow choice of their *x* coordinate for $-3 < x < 0$
	dM1	For substituting $x = "-\frac{4}{3}"$ into $y = x(x+3)(x-2)$ Allow substitution of their *x*, dependent on previous method mark
	A1	$A = \left(-\frac{4}{3}, \frac{200}{27}\right)$ students may list $x = -\frac{4}{3}, y = \frac{200}{27}$

Type MS · Factorising and solving a cubic

Answer 22 of 22 · Question 9 part c · 5 marks

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Question number	Scheme	Marks
	→ → 3 → 3 → 1 → 1	